

Machine Learning Task for TDT4173

SINTEF Energy Research

Task Overview

The project focuses on a component of short-term hydropower scheduling called the Unit Commitment (UC) problem. UC determines whether each generating unit should be turned on or off at each hour over a scheduling horizon. This is an important part of planning hydropower production.

Short-term hydropower scheduling is challenging because hydropower systems contain many reservoirs, power plants, and operational constraints. The input data are large and complex, with detailed information across time and locations. Water stored in reservoirs can be used now or saved for later, creating trade-offs between current production and future opportunities. In addition, water released from an upstream reservoir affects downstream power plants. These dependencies make the scheduling problem difficult to solve.

Traditional hydropower scheduling software uses mathematical optimization methods. These methods must decide both the on/off status of generating units and their production levels. Solving these optimization problems is computationally expensive. In addition, a solution for one scheduling case cannot be directly reused for another, so the optimization must be repeated each time.

The InterOpt research project at SINTEF Energy combines machine learning with optimization. The goal is to train a machine learning model to predict the on/off status of generating units from input data. These predictions can then help speed up and improve the optimization process.

The study uses data from the Tokke–Vinje hydropower system in southern Norway. The system contains 17 reservoirs, 9 power plants, and 14 generating units, with a total installed capacity of 1000 MW and annual production of about 4.9 TWh. Each scheduling case covers one week (168 hours). The dataset contains daily rolling-horizon cases from 2015-01-01 to 2024-12-25, giving a total of 3,647 cases.

ML Task

In this project, you should design and train a machine learning model that:

- takes time-series inputs such as electricity price, reservoir inflow, storage level, and water value
- optionally uses additional information such as environmental constraints, watercourse topology, and system configuration

- outputs a value between 0 and 1 for every generating unit and every hour. A value close to 1 indicates high confidence that the unit is ON, while a value close to 0 indicates high confidence that the unit is OFF. For example, a prediction of 0.9 means the model believes there is a high likelihood that the unit will be operating during that hour.

Dataset

The dataset consists of two parts (see Dataset_definitions_and_explanation.docx for details):

The core dataset includes:

- hourly market prices
- hourly reservoir inflows
- daily reservoir storage levels
- daily water values
- ground-truth UC decisions for each generating unit at each hour

The extended dataset includes optional information that provides background information on the watercourse, such as:

- watercourse topology
- physical system configuration
- hourly minimum reservoir volumes
- hourly minimum river flow requirements

Evaluation

You will submit the predicted probabilities of the UC states in the test set through Kaggle. Your predictions will be compared with the hidden ground-truth UC decisions. Performance will be evaluated using the ROC-AUC metric (with micro average).